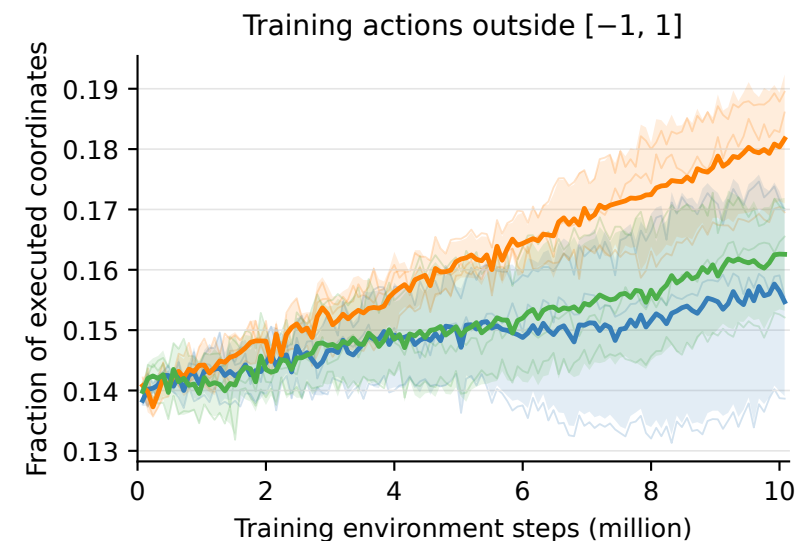
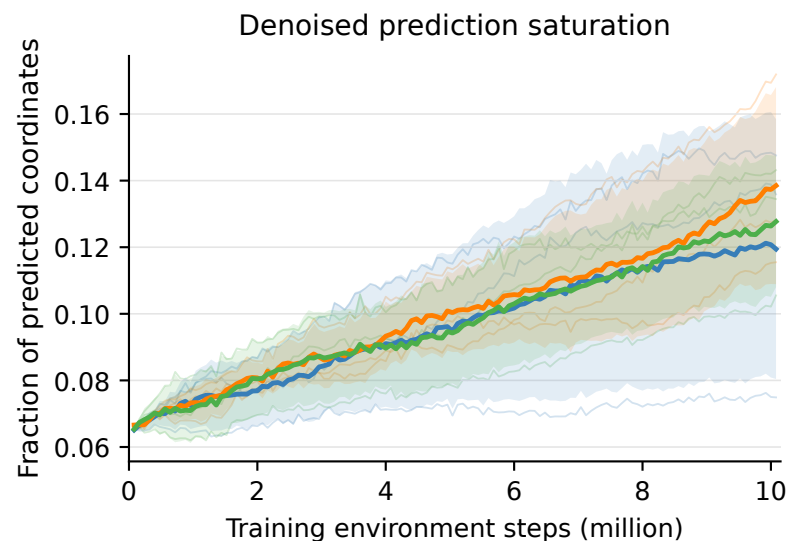
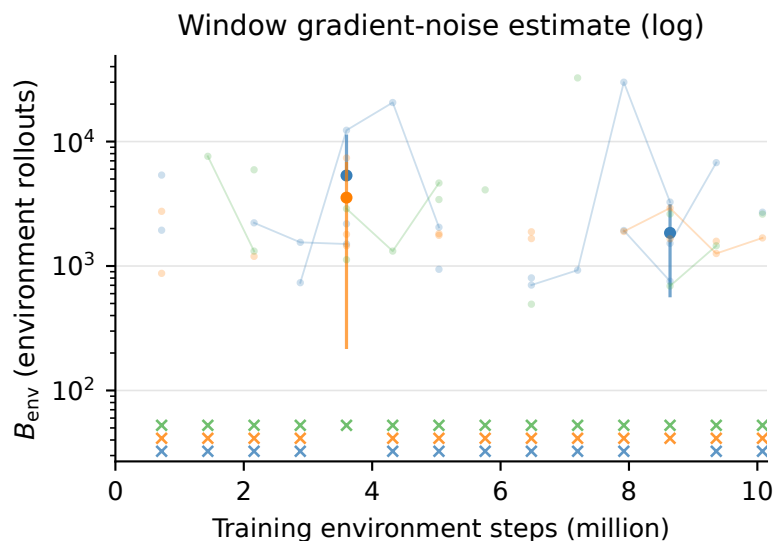


Stage 3: gradient noise and saturation across completed wave 1

Nine conditions · seeds 0-2 · mean $\pm$ sample SD · faint individual seeds · no smoothing

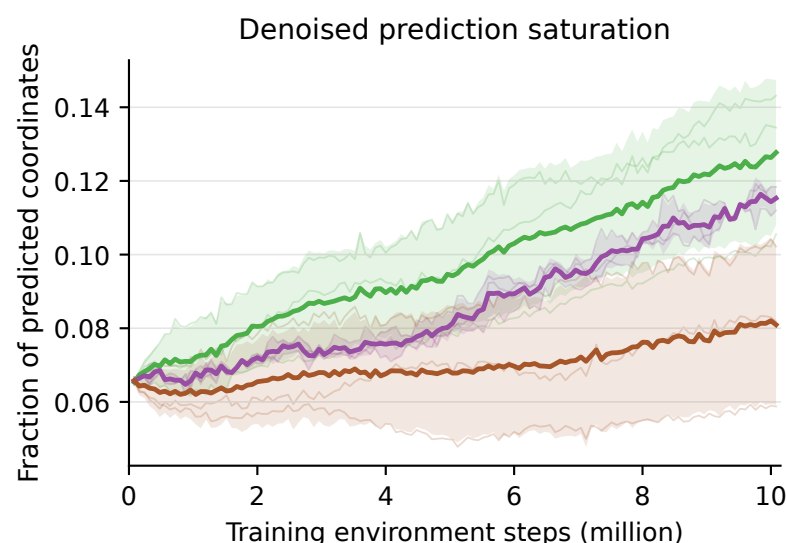
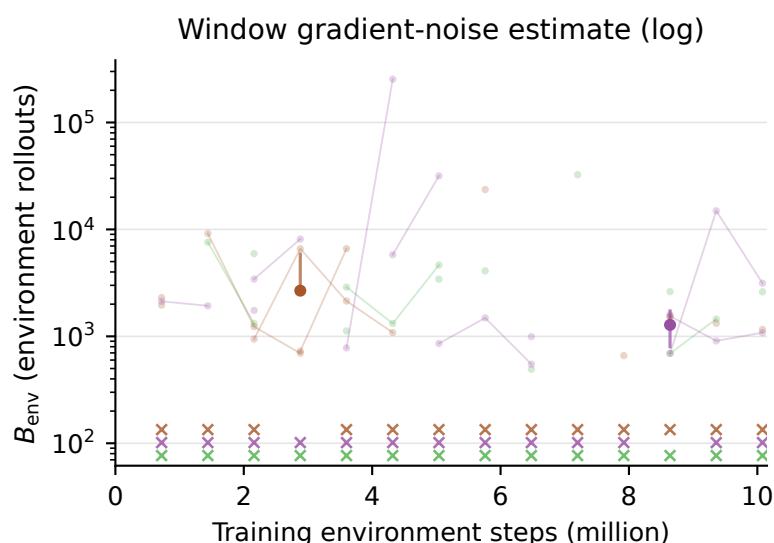
Estimator ladder under AdamW

— E0 (2/14) — E1 (1/14) — E2 (0/14)



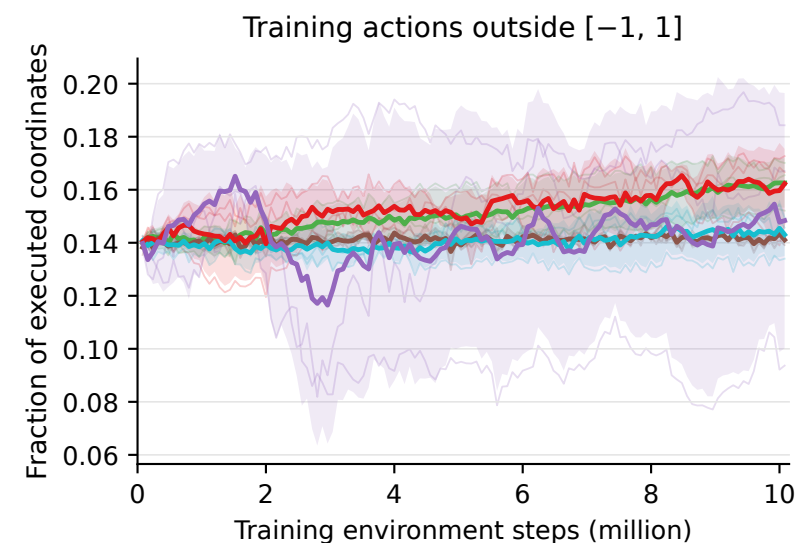
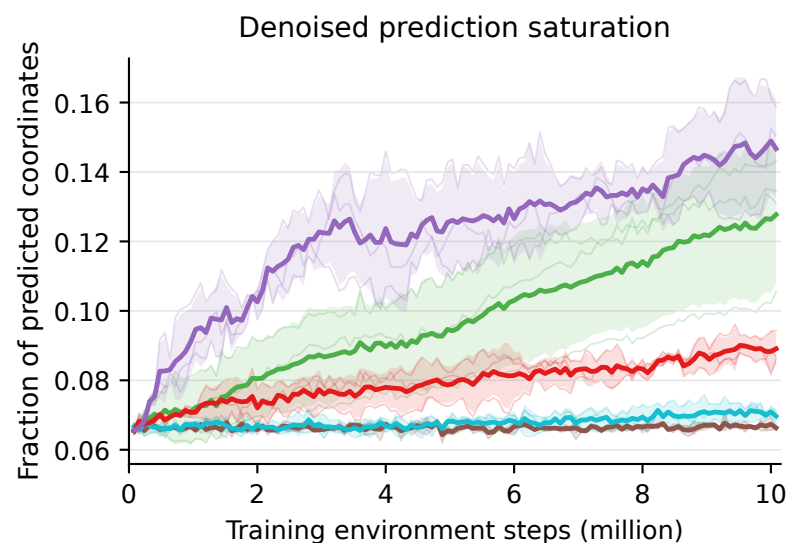
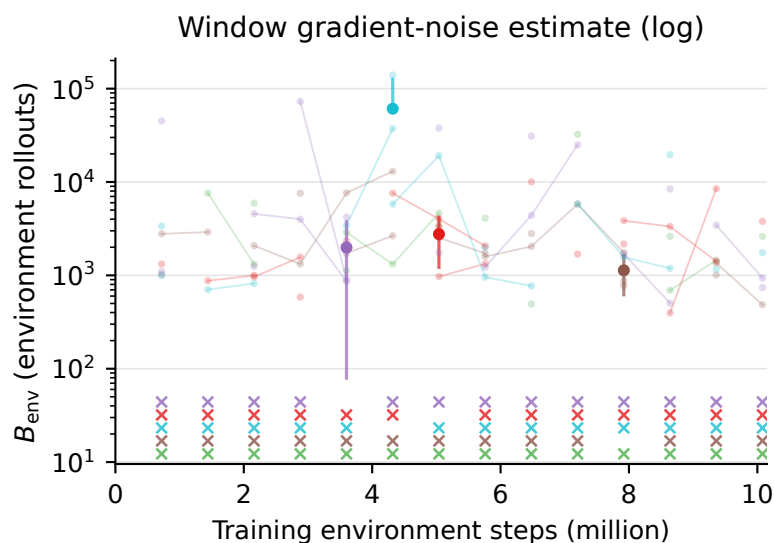
E2 under AdamW: momentum and baseline

— E2 (0/14) — E2_beta0 (1/14) — E2_nobase (1/14)



E2 estimator: AdamW and the SGD step-size sweep

— E2 (0/14) — SGD03 (1/14) — SGD1 (1/14) — SGD3 (1/14) — SGD10 (1/14)



B_{env} crosses: no three-seed mean; legend counts: defined means / 14 windows. Cross height has no data meaning.

Dots/lines retain valid individual estimates and gaps; no averaging across only the remaining positive-denominator seeds.